



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Powered Smart Finance Management System	
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Abstract

The AI-Powered Smart Finance Budget Management System is a full-stack web application developed to help users efficiently manage personal finances by recording income, tracking expenses, setting budgets, and analyzing financial activities. The system is developed using React, Spring Boot, Hibernate, JPA, MySQL, JWT authentication, and Docker, providing a secure, scalable, and user-friendly platform for financial management.

To enhance the system beyond traditional expense tracking, Artificial Intelligence and Machine Learning are integrated to analyze historical transaction data, identify spending patterns, predict future expenses, detect unusual spending behavior, and provide personalized budget recommendations. The AI component follows an MLOps lifecycle involving data preprocessing, model training, evaluation, experiment tracking using MLflow, and model version management using DVC.

The project also follows Adaptive Software Engineering principles by applying Agile and Scrum practices, including user stories, product backlog, sprint planning, and incremental development. DevOps practices are implemented using Git, GitHub, Docker, Docker Compose, CI/CD, and Kubernetes. Security is incorporated through JWT, OTP-based email verification, role-based access control, SonarQube for code analysis, OWASP ZAP for web security testing, and Trivy for container vulnerability scanning.

Overall, the proposed system combines full-stack development, AI/ML, Agile methodology, DevOps, SecOps, and MLOps to create an intelligent and secure financial management platform that helps users make better budgeting and spending decisions.

Signature of the Guide

HOD